Multi-Digit Multiplication: Partial Products & Ratio Tables

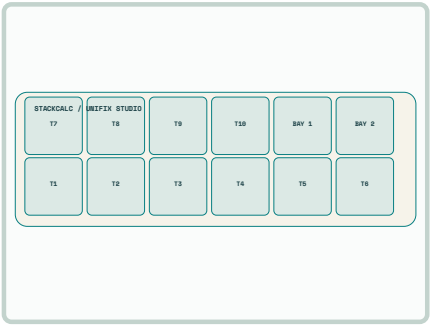
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.4.NBT.B.5

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Unifix Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Model 4 × 23 in the Array & Base-Ten Studio as 4 rows of 20 and 4 rows of 3.

Sketch Model on Paper:
Draw an open array rectangle split into 20 and 3, height 4.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	4	X: 4	Multiplier 4 into X
2.	ENTER	Y: 4 | X: 4	Stack lift
3.	20	Y: 4 | X: 20	Tens part (20) into
4.	×	X: 80	1st partial product:

EXPECTED LCD X: 92

1 **Calculate 3 × 42 using partial products: (3×40) + (3×2).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Calculate 5 × 34 using partial products: (5×30) + (5×4).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Calculate 6 × 52 using partial products: (6×50) + (6×2).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **A box holds 25 markers. How many in 8 boxes?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 1

Multi-Digit Multiplication: Partial Products & Ratio Tables • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Counting & Array Studio (COUNT-01)

CAD Blueprint: unifix_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/unifix_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Charcoal Array Tray & Teal Counting Cubes (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Calculate 3 × 42 using partial products: (3×** Standard: CCSS.MATH.4.NBT.B.5	**126**	[3] [ENTER] [4][0] [×] [3] [ENTER] [2]
2	**Calculate 5 × 34 using partial products: (5×** Standard: CCSS.MATH.4.NBT.B.5	**170**	[5] [ENTER] [3][0] [×] [5] [ENTER] [4]
3	**Calculate 6 × 52 using partial products: (6×** Standard: CCSS.MATH.4.NBT.B.5	**312**	[6] [ENTER] [5][0] [×] [6] [ENTER] [2]
4	**A box holds 25 markers. How many in 8 boxes?** Standard: CCSS.MATH.4.NBT.B.5	**200 markers**	[8] [ENTER] [2][5] [×]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Ensure students multiply the tens place value (20), not just the digit 2.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.